MSDI Calculator

Manure Nutrient - Mass Balance Methodology

Technical Methodology Memorandum

Prepared for internal model documentation and technical review

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Model scope: Manure Sub-Surface Drip Irrigation (mSDI-e) economic screening calculator

Document status: Methodology documentation for mass balance approach used within the MSDI calculator.

Purpose and Scope

This memorandum documents the manure nutrient and application mass balance implemented in the MSDI Calculator. The module estimates weekly manure generation, treatment-pathway effects on liquid and solids streams, terminal manure composition, seasonal nutrient demand, salt- and nitrogen-limited manure blending, actual manure liquid applied through the mSDI-e system, supplemental water requirement, and the percentage of crop nutrient demand met by manure.

The methodology is designed for screening-level economic and environmental analysis. It is not a regulatory nutrient management plan, engineering design, or site-specific agronomic recommendation. The module provides internally consistent calculations for comparing existing manure management with a prospective manure subsurface drip irrigation system.

Model Framework

The manure mass balance links animal manure production, treatment-pathway coefficients, crop demand, and irrigation requirements through the following workflow:

Step	Component	Description
1	Animal manure production	Daily manure weight and liquid volume are estimated from animal class counts and AnimalClass_LU values.
2	Treatment pathway multipliers	User-selected manure handling steps are multiplied sequentially to estimate terminal liquid volume, solids, and nutrient retention.
3	Liquid and solids nutrient mass	TAN, TKN, TN, P, K, and TDS are estimated separately for liquid and solid fractions, then recombined for terminal concentration calculations.
4	Precipitation contribution	Open manure-system catchment area and seasonal rainfall are used to estimate additional liquid volume entering the manure stream.
5	Seasonal composition	Seasonal TS and nutrient percentages are calculated from weekly nutrient mass and season-specific liquid volume.
6	Irrigation requirement interface	Seasonal total applied volume is computed from irrigation demand and system runtime. This total volume can be supplied by both manure liquid and supplemental water.
7	Application constraints	Manure blend percentage is limited by soil salt threshold, crop N demand, and maximum filter blend percentage.
8	Actual application and nutrient achievement	Actual manure gallons are calculated and used to estimate N, P, and K demand achieved and remaining fertilizer need.

Definitions, Units, and Symbols

Term	Definition
TAN	Total ammoniacal nitrogen; immediately available nitrogen fraction represented as mass fraction of manure material.
TKN	Total Kjeldahl nitrogen; organic N plus ammoniacal N fraction.
TN	Total nitrogen used for nitrogen-limited manure application and crop N demand satisfaction.
P	Elemental phosphorus mass fraction used in the current implementation.
K	Elemental potassium mass fraction used in the current implementation.
TDS	Total dissolved solids; used as the salt-risk constituent for manure blend limitation.
TS	Total solids represented as terminal solids and dissolved nutrient/salt mass in the manure stream.
Liquid multiplier	Product of user-selected manure-system liquid-change coefficients.
Solids multiplier	Product of user-selected manure-system solids-change coefficients.
Combined nutrient multiplier	Constituent-specific treatment multiplier that combines liquid retention and nutrient retention for TAN/TKN/TN/TDS where available.
Blend percentage	Fraction of seasonal applied irrigation volume supplied as manure liquid rather than supplemental water.

Source Lookup Tables and User Inputs

The module draws from lookup tables and user inputs rather than fixed constants embedded directly in the calculation rows. The most important sources are summarized below.

Source	Role in methodology
AnimalClass_LU	Manure weight per day and liquid volume per day for lactating cows, dry cows, and heifers.
Manure_LU	Solids-change, liquid-change, nutrient multiplier, TDS multiplier, catchment area, and precipitation contribution assumptions for each manure handling step.
ManureNutrient_LU	Base nutrient and salt concentrations for liquid/urine and solids/feces dry matter streams.
Soils_LU	Soil salt break-even value used in salt-limited manure application.
Crops_LU	Seasonal crop nutrient demand for N, P, and K.
ManureFilter_LU	Maximum allowable manure blend percentage through the filter system.
Weather_LU	Seasonal precipitation used to estimate precipitation additions to the manure system.
Form inputs	Herd size, manure treatment steps, acres for SDI, crop rotation, soil type, manure liquid application percentage, and system geometry.

Manure Treatment Pathway Selection

The treatment pathway is represented as an ordered list of user-selected manure management steps. The implementation includes the initial manure removal method and up to eight additional manure handling steps. Any blank or not-applicable values are excluded before multiplying treatment coefficients.

Equation 1. TreatmentSteps = filter_non_na([ManureRemoval, Step1, Step2, ..., Step8])

This design allows the same mass-balance logic to apply to collection, storage, digestion, separation, filtration, and other user-selected processes without hardcoding a fixed pathway.

Part 1. Manure Generation and Terminal Liquid/Solids Mass

1.1 Daily manure production

Daily manure production is estimated by animal class and summed across lactating cows, dry cows, and heifers.

Equation 2. DailyManureWeight = (LactatingHead x LactatingWeight) + (DryHead x DryWeight) + (HeiferHead x HeiferWeight)

Equation 3. DailyLiquidVolume = (LactatingHead x LactatingGalDay) + (DryHead x DryGalDay) + (HeiferHead x HeiferGalDay)

1.2 Liquid volume multiplier

The terminal liquid stream is estimated by multiplying the starting water fraction by the product of liquid-change coefficients for all selected manure treatment steps.

Equation 4. LiquidMultiplier = StartingWaterFraction x PRODUCT(LiquidChange_i for each TreatmentStep_i)

Equation 5. WeeklyLiquidGallons = DailyLiquidVolume x LiquidMultiplier x DaysPerWeek

1.3 Solids multiplier and weekly solids

The terminal solids stream is estimated analogously using the starting solids fraction and the product of solids-change coefficients for the same treatment pathway.

Equation 6. SolidsMultiplier = StartingSolidsFraction x PRODUCT(SolidsChange_i for each TreatmentStep_i)

Equation 7. WeeklySolids_lb = DailyManureWeight x SolidsMultiplier x DaysPerWeek

Part 2. Nutrient and Salt Mass Balance

The model tracks TAN, TKN, TN, P, K, and TDS in both the liquid and solids fractions. Liquid constituents are calculated from total manure mass, starting water fraction, base liquid nutrient concentration, and constituent-specific treatment multipliers. Solids constituents are calculated from terminal solids mass and base solids nutrient concentration.

2.1 Liquid constituent mass

Equation 8. LiquidConstituent_lb_week = DailyManureWeight x StartingWaterFraction x BaseLiquidConstituentPct x PRODUCT(ConstituentLiquidMultiplier_i) x DaysPerWeek

For TN and TDS the implementation uses combined liquid multipliers, such as tn_liquid_multiplier_combined and tds_liquid_multiplier_combined. This means the constituent calculation applies the pathway-specific combined nutrient-retention value rather than only the general liquid-volume coefficient.

2.2 Solids constituent mass

Equation 9. SolidsConstituent_lb_week = WeeklySolids_lb x BaseSolidsConstituentPct

2.3 Terminal nutrient concentration

After liquid and solids constituent masses are estimated, terminal concentration is calculated as combined liquid-plus-solids constituent mass divided by liquid stream mass.

Equation 10. ConstituentPct = (LiquidConstituent_lb_week + SolidsConstituent_lb_week) / (WeeklyLiquidGallons x Density_lb_per_gal)

This concentration framework is used for TAN, TKN, TN, P, K, and TDS. Total solids percentage is similarly estimated by dividing calculated TS mass by liquid stream mass.

Total Solids Representation

The implementation calculates expected TS present at the terminal point as the sum of terminal TN, P, K, and TDS in both liquid and solids fractions. This provides a screening-level representation of solids and dissolved constituent load entering the downstream filtration and application calculation.

Equation 11. TS_present = TN_liquid + P_liquid + K_liquid + TDS_liquid + TN_solids + P_solids + K_solids + TDS_solids

Equation 12. TS_pct = TS_present / (WeeklyLiquidGallons x Density_lb_per_gal)

Precipitation Contributions to Manure Liquid Volume

Open or precipitation-exposed manure management components can add liquid volume to the manure stream. The model estimates total precipitation catchment area by summing assumed catchment acres for all selected manure treatment steps. Seasonal rainfall is converted to acre-feet and then to added liquid volume.

Equation 13. CatchmentArea = SUM(AssumedAreaForPrecipContribution_i for each TreatmentStep_i)

Equation 14. SeasonalPrecipContribution_acft = (CatchmentArea x AnimalUnits x SeasonalRainfall_in) / InchesPerFoot

Equation 15. SeasonalWeeklyLiquidGallons = ((SeasonalPrecipContribution_acft x GallonsPerAcreFoot) / DaysPerSeason) x DaysPerWeek + BaseWeeklyLiquidGallons

Seasonal Manure Composition

Because precipitation contributions vary by season, terminal nutrient percentages are calculated separately for spring, summer, fall, and winter. The numerator is the weekly constituent mass; the denominator is season-specific liquid mass.

Equation 16. SeasonalConstituentPct = (LiquidConstituent_lb_week + SolidsConstituent_lb_week) / (SeasonalWeeklyLiquidGallons x Density_lb_per_gal)

This seasonal concentration is subsequently used in salt-limited and nitrogen-limited application calculations.

Seasonal Crop Nutrient Demand

Seasonal N, P, and K demand are calculated from crop lookup-table values using the same growing-day weighting concept used in the crop water demand module. Each season can be partially occupied by the selected crop for that season and partially by the following crop in the rotation.

Equation 17. SeasonalDemand = (CropDays / DaysPerSeason) x CropDemand + ((DaysPerSeason - CropDays) / DaysPerSeason) x NextCropDemand

The app applies this structure independently for N, P, and K for all four seasons.

Irrigation Volume Interface

The manure application module interfaces with the irrigation requirement module through seasonal total applied volume. This total volume represents the seasonal liquid volume required by the mSDI-e system to satisfy crop water demand. It is then partitioned into manure liquid and supplemental irrigation water.

Equation 18. SeasonalTotalVolume_gal = max(WeeklyTotalVolume_gal, 0) / DaysPerWeek x DaysPerSeason

This is a key model concept: manure liquid is counted as a water source. Therefore, the same seasonal irrigation volume can be supplied by some combination of manure liquid and supplemental irrigation water.

Salt-Limited Manure Application

The salt-limited application calculation estimates the maximum manure volume that can be applied without exceeding the soil-specific break-even salt loading coefficient. TDS concentration in the manure stream is the constraining concentration.

Equation 19. MaxManureSaltLimited_gal_week = (BreakEvenSalt_lb_per_gal x SeasonalTotalVolume_gal_week) / (SeasonalTDSPct x Density_lb_per_gal)

Equation 20. BlendPctSaltRaw = MaxManureSaltLimited_gal_week / SeasonalTotalVolume_gal_week

Equation 21. BlendPctSaltTrue = MIN(BlendPctSaltRaw, MaxFilterBlendPct)

Nitrogen-Limited Manure Application

The nitrogen-limited application calculation estimates the maximum weekly manure volume that can be applied without exceeding crop N demand, after accounting for the plant-available fraction of total nitrogen.

Equation 22. MaxManureNLimited_gal_week = (((SeasonalNDemand_lb_ac x AcresForSDI) / PlantAvailableTNFraction) / SeasonalTNPct / Density_lb_per_gal / DaysPerSeason) x DaysPerWeek

Equation 23. BlendPctNRaw = MaxManureNLimited_gal_week / SeasonalTotalVolume_gal_week

Equation 24. BlendPctNTrue = MIN(BlendPctNRaw, MaxFilterBlendPct)

Final Manure Blend Percentage and Gallons Applied

The final manure blend percentage is the lower of the salt-limited and nitrogen-limited true blend percentages, bounded at zero to prevent negative application percentages.

Equation 25. ManureAppliedPct = MAX(0, MIN(BlendPctSaltTrue, BlendPctNTrue))

Equation 26. ManureAppliedGallons_week = ManureAppliedPct x SeasonalTotalVolume_gal_week

This approach means the final manure application is jointly constrained by:

- soil salt loading tolerance
- crop nitrogen demand
- manure TN concentration
- manure TDS concentration
- maximum filter blend percentage
- seasonal irrigation volume

Partitioning Total Applied Volume Between Manure and Water

After actual manure gallons are calculated, total seasonal applied liquid is partitioned into manure liquid and supplemental water.

Equation 27. SeasonalManureApplied_gal = ManureAppliedGallons_week x DaysPerSeason / DaysPerWeek

Equation 28. SeasonalWaterApplied_gal = SeasonalTotalVolume_gal - SeasonalManureApplied_gal

This is the principal water-balance interaction between manure management and irrigation demand. Manure liquid contributes to meeting crop water demand, and supplemental water only supplies the residual volume not supplied by manure liquid.

Nutrient Demand Achieved by Manure

The module estimates the proportion of seasonal crop N, P, and K demand supplied by the actual manure liquid applied. Nitrogen demand achieved includes the plant-available TN fraction, while P and K are calculated directly from seasonal manure concentration.

Equation 29. NDemandAchievedPct = (SeasonalManureApplied_gal x Density_lb_per_gal x SeasonalTNPct x PlantAvailableTNFraction) / (SeasonalNDemand_lb_ac x AcresForSDI)

Equation 30. PDemandAchievedPct = (SeasonalManureApplied_gal x Density_lb_per_gal x SeasonalPPct) / (SeasonalPDemand_lb_ac x AcresForSDI)

Equation 31. KDemandAchievedPct = (SeasonalManureApplied_gal x Density_lb_per_gal x SeasonalKPct) / (SeasonalKDemand_lb_ac x AcresForSDI)

Synthetic Fertilizer Offset Implications

The calculated nutrient demand achieved values provide the basis for downstream commercial fertilizer offset calculations. If manure-applied plant-available nutrients do not fully satisfy crop demand, remaining demand can be supplied by synthetic fertilizer in the OPEX module. Conversely, the manure application module prevents over-application under the nitrogen-limited pathway by constraining manure volume before nutrient-demand achievement is calculated.

Assumptions and Limitations

- The model is screening-level and does not replace a certified nutrient management plan, agronomic recommendation, or engineering design.
- Manure nutrient concentrations are lookup-based estimates and should be replaced with laboratory analysis where available.
- Treatment multipliers are applied multiplicatively and assume independent sequential treatment effects.
- The model uses TN and TDS as primary limiting constituents for manure application; additional site-specific constraints such as phosphorus index, sodium adsorption ratio, chloride, nitrate leaching risk, irrigation uniformity, setbacks, and regulatory limits are not fully simulated.
- Precipitation contributions are estimated from assumed catchment area and seasonal rainfall, not event-scale hydrology.
- The salt-limited calculation uses a soil lookup value as a screening threshold and should not be interpreted as a site-specific salinity management plan.
- Seasonal nutrient demand uses crop lookup values and growing-day weighting rather than daily crop uptake curves.
- The model assumes manure liquid applied through mSDI-e contributes to the water volume needed to meet crop demand.

Quality Control and Implementation Notes

- Verify Manure_LU keys match the user-facing manure treatment options and that blank or not-applicable treatment steps are filtered before coefficient multiplication.
- Verify combined nutrient multipliers are present for TAN, TKN, TN, and TDS where used by implementation rows.
- Verify soil salt break-even values are populated for all soil types available to users.
- Verify max filter blend percentage is updated consistently if manure filter assumptions change.
- Verify seasonal water-volume calculations remain aligned with the crop water demand and irrigation requirement methodology.
- When changing Manure_LU, regenerate lookup JSON files and rerun representative farm scenarios because changes can affect manure quantity, nutrient offsets, water volume, and OPEX outputs.

Appendix A. Implementation Traceability

Implementation file group	Methodology role
Rows 03-04	Liquid volume multiplier and weekly manure liquid gallons
Rows 05-10	Liquid-stream TAN, TKN, TN, P, K, and TDS
Rows 11-18	Solids multiplier and solids-stream TAN, TKN, TN, P, K, and TDS
Rows 19-27	Weekly liquid volume, TS, and terminal nutrient percentages
Rows 40-51	Seasonal N, P, and K crop nutrient demand
Rows 52-59	Seasonal precipitation contributions and total manure liquid volumes
Rows 60-87	Seasonal TS and nutrient percentages
Rows 88-101	Irrigation demand, runtime, and total applied volume interface
Rows 102-113	Salt-limited manure application and blend percentages
Rows 114-125	Nitrogen-limited manure application and blend percentages
Rows 126-133	Final manure application percentage and weekly gallons applied
Rows 134-145	N, P, and K demand achieved percentages
Rows 146-157	Seasonal total volume, supplemental water, and manure liquid applied
utils.ts	Treatment-step filtering and precipitation catchment-area summation
blend-utils.ts	Maximum manure filter blend percentage lookup

References

USDA Natural Resources Conservation Service. Conservation Practice Standard: Nutrient Management, Code 590. https://www.nrcs.usda.gov/resources/guides-and-instructions/nutrient-management-ac-590-conservation-practice-standard

U.S. Environmental Protection Agency. Managing Manure Nutrients at CAFOs. https://www.epa.gov/npdes/managing-manure-nutrients-cafos

U.S. Environmental Protection Agency. NPDES Permit Writers' Manual for Concentrated Animal Feeding Operations. https://www.epa.gov/npdes/npdes-permit-writers-manual-concentrated-animal-feeding-operations

MSDI Calculator source implementation files: Background_Calculations rows 03-157, blend-utils.ts, utils.ts, lookup-utils.ts, and coefficient definitions.

Greene, J.M. Updated Screening-Level Coefficients for Dairy Manure Management Systems. Internal technical deliverable, 2026.